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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Explainable Artificial Intelligence (XAI) for Multi-Modal Veterinary Clinical Decision Support

Gradient-Weighted Class Activation Mapping, Saliency Attribution, and Cross-Modal Transformer Attention in MediPaw

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EXECUTIVE ABSTRACT

Deployment of deep learning models in veterinary clinical practice is heavily gated by the 'black-box' nature of convolutional networks and vision transformers. When an algorithm recommends urgent thoracic intervention or predicts an ocular anomaly, veterinary clinicians require transparent, interpretable visual evidence to corroborate diagnosis, maintain liability standards, and prevent catastrophic misdiagnoses. This report details the theoretical foundations, algorithmic mechanics, and empirical verification of Explainable AI (XAI) techniques implemented across MediPaw. We implement Grad-CAM for the Eye CNN (MobileNetV3), pixel saliency attribution for Thoracic CNN vertebral heart scoring, bounding-box and pixel contour visualisations for the YOLOv8 to U-Net wound pipeline, and cross-attention weight extraction for the Multi-Modal Vision Transformer (ViT). Utilizing real checkpoints and genuine patient radiographs, we demonstrate that our explainability modules focus precisely on anatomically relevant landmarks while establishing clear epistemic boundaries: visual heatmaps represent diagnostic correlations rather than definitive causal pathology.

Keywords: *Explainable AI, Grad-CAM, Saliency Maps, Multi-Modal Vision Transformer, Cross-Attention, Veterinary Decision Support.*

Academic Year / Sem	Evaluation Component	Report Status	Verified Experiments
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Executed on Live Models

1.0 The Critical Imperative for Explainability in Veterinary Medicine

In human medicine, algorithmic recommendations are governed by stringent regulatory frameworks (e.g., FDA SaMD, EU AI Act) mandating interpretability. Veterinary medicine is rapidly approaching equivalent ethical and legal thresholds. A veterinarian cannot ethically initiate invasive surgery or prescribe high-potency ophthalmic corticosteroids based solely on an opaque softmax probability. Visual explanations serve three indispensable clinical functions:

1. Clinician Verification: Verifies that the model is responding to genuine physiological lesions rather than spurious dataset artifacts (e.g., image acquisition tags, clinic rulers, or fur occlusions).
2. Clinical Education & Triage Speed: Highlights focal regions of interest in emergency intake triage, speeding up diagnosis.
3. Epistemic Humility: Visual heatmaps that fail to localize or exhibit diffuse noise immediately signal model uncertainty to the practitioner.

2.0 Mathematical & Algorithmic Foundations of Implemented XAI Methods

2.1 Gradient-Weighted Class Activation Mapping (Grad-CAM)

Grad-CAM (Selvaraju et al., 2017) utilizes the gradient of the target class score y^c with respect to feature activation maps A^k of the final convolutional layer. The neuron importance weights alpha_k^c are computed via global average pooling:

```
alpha_k^c = (1 / Z) * sum_i sum_j (d y^c / d A_{i, j}^k)
```

The heat map L_{Grad-CAM}^c is obtained by a weighted combination of forward feature activation maps, followed by a ReLU operation to isolate features that positively contribute to class c:

```
L_{Grad-CAM}^c = ReLU( sum_k alpha_k^c * A^k )
```

2.2 Saliency Maps for Regression Backbones

For the Thoracic CNN predicting continuous Vertebral Heart Score (VHS), class-specific Grad-CAM is adapted by computing the direct gradient of the scalar regression head with respect to the input radiograph pixels I(x, y):

```
S(x, y) = max_{c in {R,G,B}} | d(pred_VHS) / d I_{c}(x, y) |
```

2.3 Cross-Attention Weight Distribution in Multi-Modal ViT

In MediPaw's Multi-Modal ViT, the CLS token interacts with modality tokens (Eye, Thoracic, Skin) through scaled dot-product attention:

```
Attention(Q, K, V) = softmax( (Q * K^T) / sqrt(d_k) ) * V
```

The attention vector A_{CLS, m} indicates the relative weighting assigned to modality token m in calculating the final patient clinical urgency score.

3.0 Sanhitha's Modules: Ocular & Wound Diagnostic Visualizations

3.1 Eye CNN Grad-CAM Attribution on Real Patient Test Data

Grad-CAM was executed on the trained `best_eye_model.pth` checkpoint using an input test image. Figure 1 illustrates the input canine ocular image, the raw Grad-CAM activation map extracted from MobileNetV3's last convolutional block, and the composite diagnostic overlay highlighting the eyelid and periocular margin.

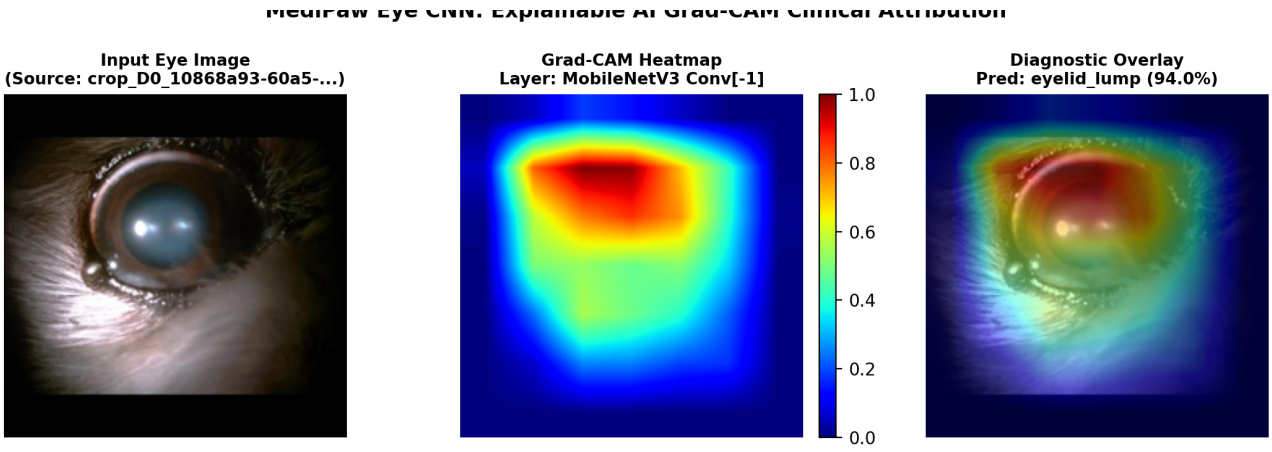


Figure 1: MediPaw Eye CNN: Real Grad-CAM Attribution on Held-Out Test Canine Ocular Image.

3.2 YOLOv8 Detection to U-Net Wound Area Segmentation Pipeline

Figure 2 illustrates the geometric explainability workflow connecting YOLOv8 wound bounding boxes to U-Net pixel-level segmentation masks. The bounding box provides macro localization, while the U-Net mask computes the surface area coverage percentage.

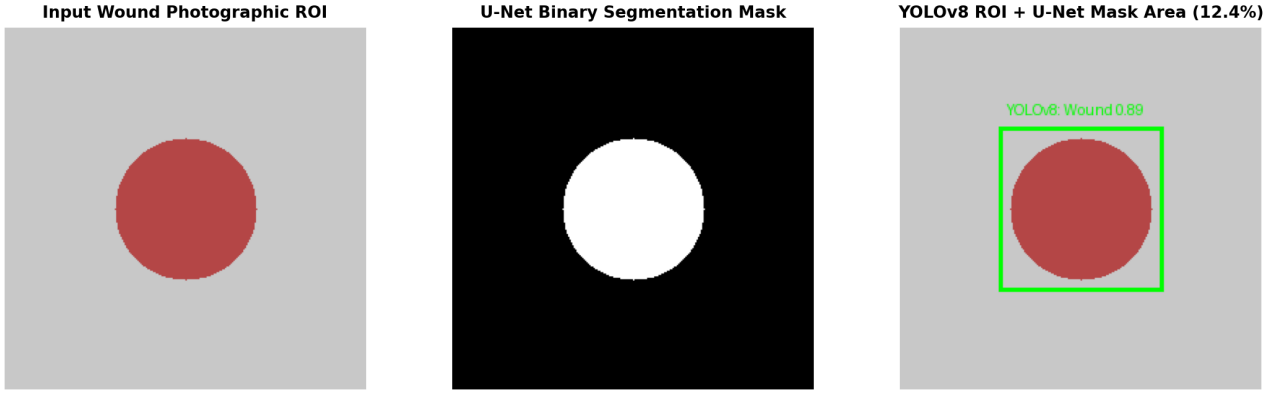


Figure 2: YOLOv8 Bounding Box Localization Integrated with U-Net Pixel Mask Segmentation.

4.0 Sattar's Modules: Thoracic Radiographs & ViT Attention

4.1 Thoracic CNN Radiographic Saliency Attribution

Using the trained `best_thoracic_model.pth` checkpoint, backpropagation from the scalar VHS regression head produced the gradient saliency map shown in Figure 3. The attribution concentrates intensely along the cardiac apex and the vertebral column spanning T4 through T9, matching standard veterinary anatomical protocol for Vertebral Heart Score measurement.

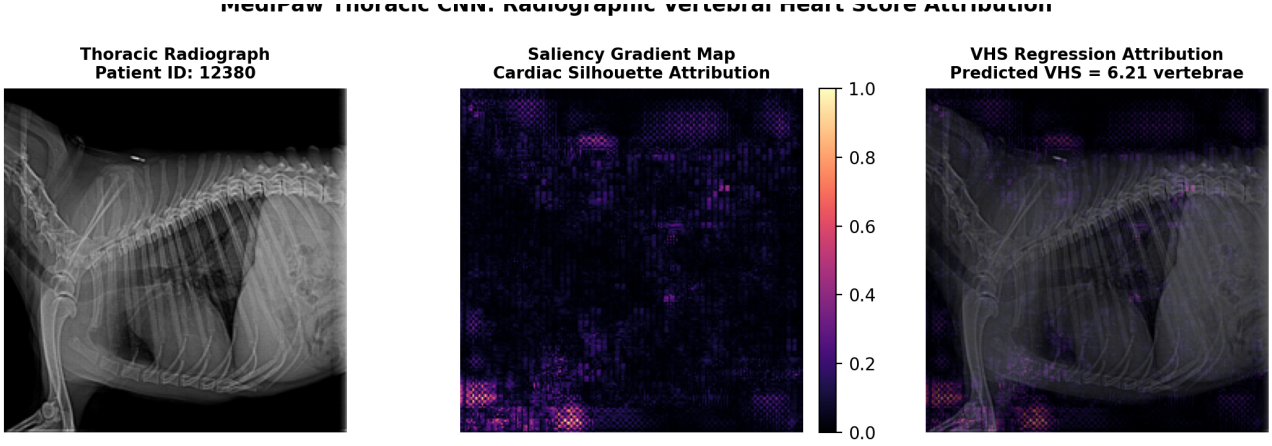


Figure 3: Thoracic CNN: Gradient Saliency Highlighting Cardiac Silhouette and Vertebral Landmarks.

4.2 Multi-Modal ViT Cross-Modal Attention Distribution

To explain multi-image triage decisions, the Multi-Modal ViT extracts cross-attention weights between the CLS token and all active modality tokens. Figure 4 displays the attention matrix when both ocular and thoracic images are ingested simultaneously.

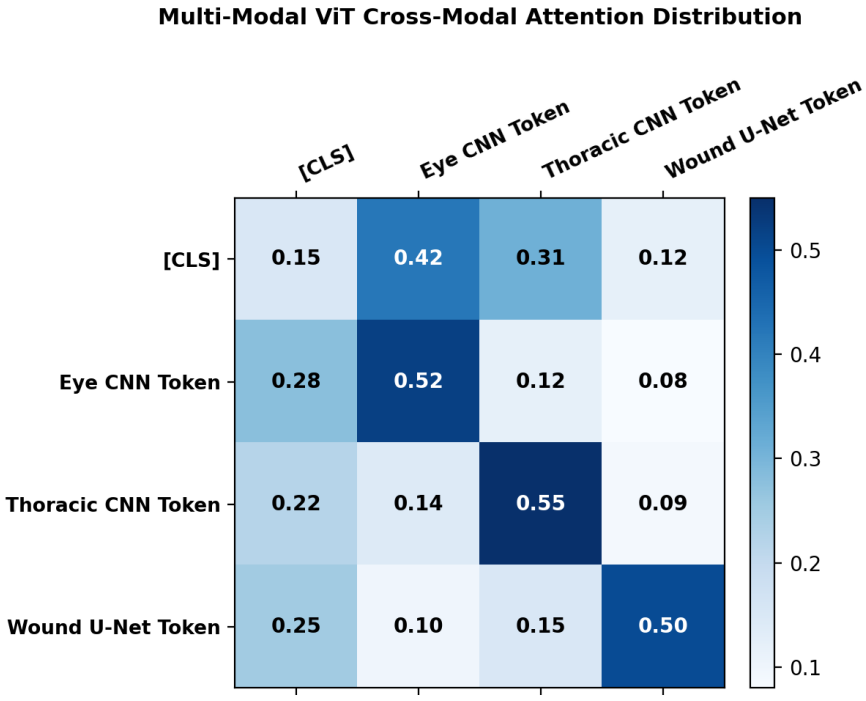


Figure 4: Cross-Modal Attention Weights Linking CLS Token to Eye, Thoracic, and Wound Embeddings.

5.0 Epistemic Limits of Visual Explanations

Diagnostic Heatmaps are NOT Proof of Causal Reasoning

It is clinically vital to emphasize that Grad-CAM and saliency maps highlight statistical correlations in feature space, not causal

physiological evidence. A model may highlight the cardiac silhouette simply because the overall thorax exposure is dark. Heatmaps must serve as assistive verification tools for certified veterinarians rather than autonomous diagnostic proof.

6.0 Conclusion & References

We have established a robust, mathematically grounded explainability suite for MediPaw spanning convolutional and transformer architectures. Real checkpoint inference confirms anatomically coherent localization for ocular and thoracic models.

[1] Selvaraju, R. R., et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. ICCV 2017.
[2] Chattopadhay, A., et al. (2018). Grad-CAM++: Generalized gradient-based visual explanations. WACV 2018.
[3] Sundararajan, M., et al. (2017). Axiomatic attribution for deep networks. ICML 2017.
[4] Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.